Lesson 1: Disorder of Hemostasis (Questions 1–20)

1. What is the main function of hemostasis?

A. Produce red blood cells

B. Stop blood loss after blood vessel injury

C. Destroy all blood clots

D. Increase blood pressure

Correct answer: B. Stop blood loss after blood vessel injury

2. Which stage of hemostasis immediately narrows an injured blood vessel to reduce blood loss?

A. Primary hemostasis

B. Secondary hemostasis

C. Vasoconstriction

D. Fibrinolysis

Correct answer: C. Vasoconstriction

3. Which substance is released by damaged endothelial cells and causes vasoconstriction?

A. Endothelin B. Plasmin C. Fibrinogen D. Albumin

Correct answer: A. Endothelin

4. What is the main role of platelets in primary hemostasis?

A. Dissolve fibrin clots B. Produce red blood cells C. Form a temporary platelet plug D. Prevent all coagulation

Correct answer: C. Form a temporary platelet plug

5. Which protein helps platelets adhere to exposed collagen at an injured vessel?

A. Albumin

B. von Willebrand factor (vWF)

C. Plasmin

D. Hemoglobin

Correct answer: B. von Willebrand factor (vWF)

6. Which substances are released by activated platelets to promote platelet activation and aggregation?

A. ADP and thromboxane A_2

B. Insulin and glucagon

C. Albumin and globulin

D. Plasmin and plasminogen

Correct answer: A. ADP and thromboxane A_2

7. What is the main goal of secondary hemostasis?

A. Destroy platelets

B. Narrow blood vessels permanently

C. Produce a strong fibrin clot

D. Break down fibrin

Correct answer: C. Produce a strong fibrin clot

8. Which enzyme converts fibrinogen into fibrin?

A. Plasmin

B. Thrombin

C. Albumin

D. Amylase

Correct answer: B. Thrombin

9. Which coagulation pathway is primarily responsible for initiating the coagulation cascade?

A. Extrinsic pathway

B. Intrinsic pathway

C. Fibrinolytic pathway

D. Platelet pathway

Correct answer: A. Extrinsic pathway

10. Which coagulation pathway is primarily involved in amplification of the coagulation cascade?

A. Extrinsic pathway

B. Intrinsic pathway

C. Fibrinolytic pathway

D. Complement pathway

Correct answer: B. Intrinsic pathway

11. What is the main function of fibrin in a blood clot?

A. Break down platelets

B. Stabilize the platelet plug

C. Reduce platelet production D. Dissolve the clot

Correct answer: B. Stabilize the platelet plug

12. Which enzyme primarily breaks down fibrin during fibrinolysis?

A. Thrombin

B. Factor VIII

C. Plasmin

D. Fibrinogen

Correct answer: C. Plasmin

13. Which substance converts plasminogen into plasmin?

A. Tissue plasminogen activator (tPA)

B. Hemoglobin

C. Albumin

D. Factor VIII

Correct answer: A. Tissue plasminogen activator (tPA)

14. What is the main purpose of fibrinolysis?

A. Form a platelet plug

B. Narrow the injured vessel

C. Remove the clot after vessel repair

D. Produce more fibrin

Correct answer: C. Remove the clot after vessel repair

15. Hemostatic disorders can be classified by which of the following?

A. Origin and affected hemostatic component

B. Blood group only

C. Body temperature only

D. Age and height only

Correct answer: A. Origin and affected hemostatic component

16. Which of the following is an inherited hemostatic disorder?

A. Vitamin K deficiency

B. Liver disease

C. Hemophilia

D. Acquired disseminated intravascular coagulation

Correct answer: C. Hemophilia

17. Which condition is an example of a disorder affecting multiple components of hemostasis?

A. Hemophilia A

B. Disseminated intravascular coagulation (DIC)

C. Glanzmann thrombasthenia

D. A simple vessel injury

Correct answer: B. Disseminated intravascular coagulation (DIC)

18. Which laboratory test assesses platelet function and vascular integrity by measuring the time needed for bleeding to stop?

A. Bleeding time

B. Prothrombin time (PT)

C. Thrombin time (TT)

D. Blood glucose test

Correct answer: A. Bleeding time

19. Which laboratory test is used to evaluate the intrinsic and common coagulation pathways?

A. Bleeding time

B. Activated partial thromboplastin time (aPTT)

C. Blood glucose test

D. Hemoglobin concentration

Correct answer: B. Activated partial thromboplastin time (aPTT)

20. What does an increased D-dimer level generally indicate?

A. No fibrin breakdown has occurred

B. Breakdown of cross-linked fibrin has occurred

C. Increased red blood cell production

D. Normal platelet adhesion only

Correct answer: B. Breakdown of cross-linked fibrin has occurred

Lesson 2: Coagulation Disorders (Questions 21–40)

21. Which clotting factor is deficient in Hemophilia A?

A. Factor V

B. Factor VIII

C. Factor IX

D. Factor X

Correct answer: B. Factor VIII

22. Which clotting factor is deficient in Hemophilia B?

A. Factor VII

B. Factor VIII

C. Factor IX

D. Factor XII

Correct answer: C. Factor IX

23. What is the inheritance pattern of Hemophilia A?

A. Autosomal dominant

B. X-linked recessive

C. Autosomal recessive

D. Mitochondrial inheritance

Correct answer: B. X-linked recessive

24. Which group is mainly affected by Hemophilia A?

A. Males

B. Females only

C. Newborn females only

D. All individuals equally

Correct answer: A. Males

25. What is the main treatment for Hemophilia A?

A. Factor IX replacement

B. Factor VIII replacement

C. Vitamin C supplementation

D. Platelet destruction

Correct answer: B. Factor VIII replacement

26. What is the main treatment for Hemophilia B?

A. Factor VIII replacement

B. Factor IX replacement

C. Insulin therapy

D. Antibiotic therapy

Correct answer: B. Factor IX replacement

27. Which laboratory test is commonly prolonged in Hemophilia A and Hemophilia B?

A. Bleeding time only

B. Prothrombin time (PT)

C. Activated partial thromboplastin time (aPTT)

D. Blood glucose test

Correct answer: C. Activated partial thromboplastin time (aPTT)

28. Which test evaluates the extrinsic and common coagulation pathways?

A. Prothrombin time (PT)

B. Bleeding time

C. Platelet count only

D. Hemoglobin test

Correct answer: A. Prothrombin time (PT)

29. What is the normal PT range stated in the lesson?

A. 3–10 minutes

B. 10–14 seconds

C. 30–40 seconds

D. 60–90 seconds

Correct answer: B. 10–14 seconds

30. What is the normal aPTT range stated in the lesson?

A. 3–10 seconds

B. 10–14 seconds

C. 30–40 seconds

D. 60–90 seconds

Correct answer: C. 30–40 seconds

31. What does thrombin time (TT) measure?

A. The time required for platelets to adhere

B. The time required for thrombin to convert fibrinogen into fibrin

C. The number of platelets in blood

D. The time required to produce red blood cells

Correct answer: B. The time required for thrombin to convert fibrinogen into fibrin

32. Which of the following is an acquired coagulation disorder?

A. Hemophilia A

B. Hemophilia B

C. Vitamin K deficiency

D. Inherited von Willebrand disease

Correct answer: C. Vitamin K deficiency

33. Which clotting factors depend on vitamin K?

A. Factors I, V, VIII, and XII

B. Factors II, VII, IX, and X

C. Factors III, V, VIII, and XI

D. Factors I, III, and V only

Correct answer: B. Factors II, VII, IX, and X

34. What causes von Willebrand disease (vWD)?

A. Deficiency or dysfunction of von Willebrand factor

B. Excessive red blood cell production

C. Deficiency of Factor IX only

D. Excessive albumin production

Correct answer: A. Deficiency or dysfunction of von Willebrand factor

35. Which is a common symptom of von Willebrand disease?

A. Easy bruising and nosebleeds

B. Increased bone density

C. High blood glucose

D. Reduced body temperature

Correct answer: A. Easy bruising and nosebleeds

36. What is one important function of von Willebrand factor?

A. It breaks down fibrin

B. It helps platelets adhere to the injured vessel wall

C. It destroys clotting factors

D. It converts plasmin into thrombin

Correct answer: B. It helps platelets adhere to the injured vessel wall

37. What is disseminated intravascular coagulation (DIC)?

A. A condition involving only low red blood cell production

B. A disorder involving widespread activation of blood clotting

C. An inherited deficiency of Factor VIII only

D. A condition involving excessive platelet production only

Correct answer: B. A disorder involving widespread activation of blood clotting

38. Why can DIC cause both thrombosis and bleeding?

A. It increases only red blood cell production

B. It consumes platelets and clotting factors while causing widespread clot formation

C. It prevents all clot formation from the beginning

D. It increases vitamin K production

Correct answer: B. It consumes platelets and clotting factors while causing widespread clot formation

39. Which laboratory findings are typical of DIC?

A. Increased platelet count and shortened PT/aPTT

B. Decreased platelet count, prolonged PT/aPTT, and increased D-dimer

C. Normal platelet count and decreased D-dimer in every case

D. Increased clotting factors and decreased fibrin breakdown

Correct answer: B. Decreased platelet count, prolonged PT/aPTT, and increased D-dimer

40. What is the most important treatment priority in DIC?

A. Treat the underlying cause

B. Give Factor VIII to every patient

C. Prevent all clot breakdown

D. Increase platelet destruction

Correct answer: A. Treat the underlying cause